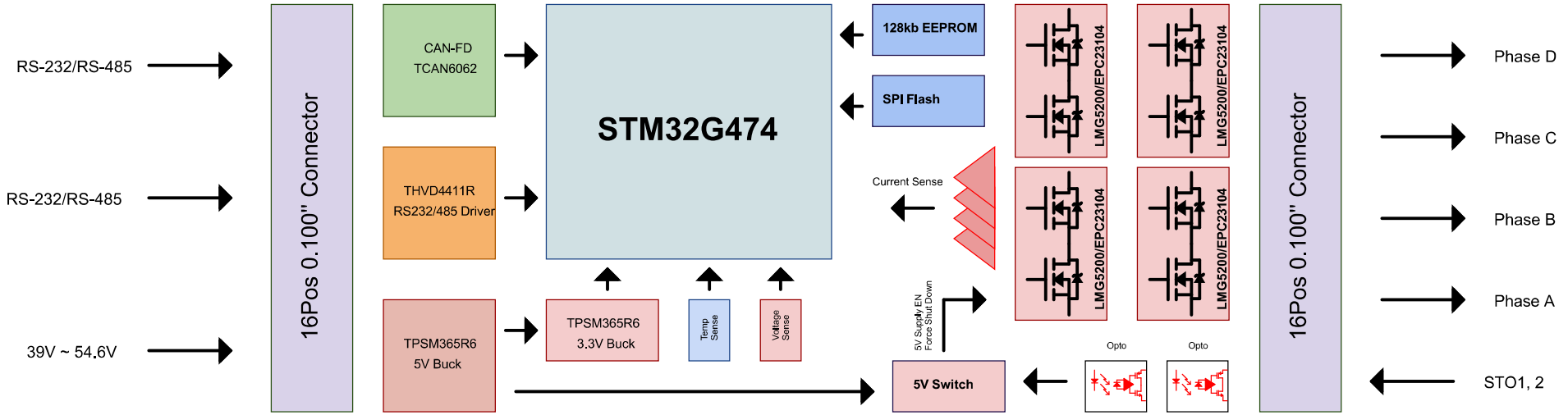
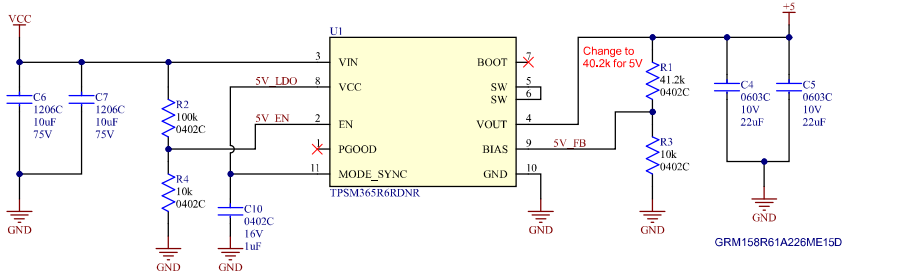
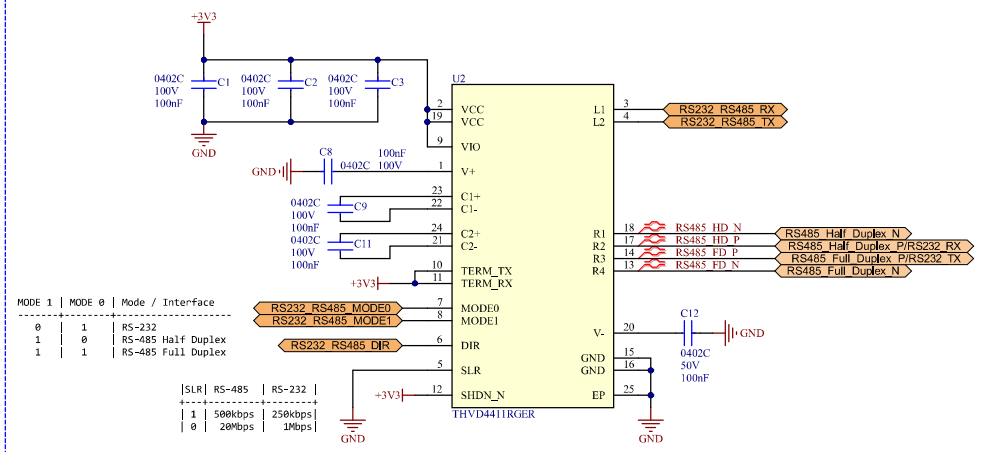


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Size: A	Number:		Revision:
Date:	9/15/2026	Sheet of	
File:	D:\Altium\GaN_SoM\test3_cover.SchDoc	Drawn By:	Melona Sawara

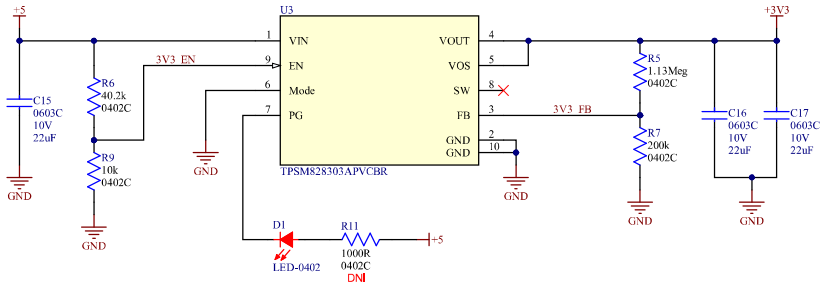
54.6V ~ 39V → 5V Buck Converter



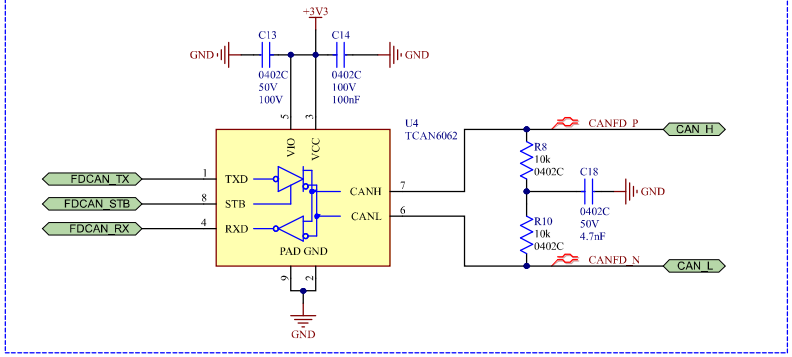
RS-232/RS-485 Driver



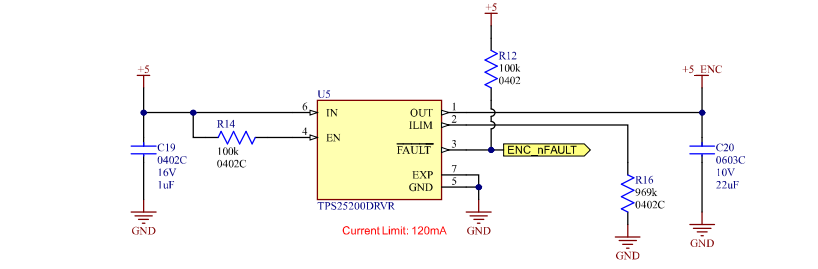
5V → 3.3V Buck Converter



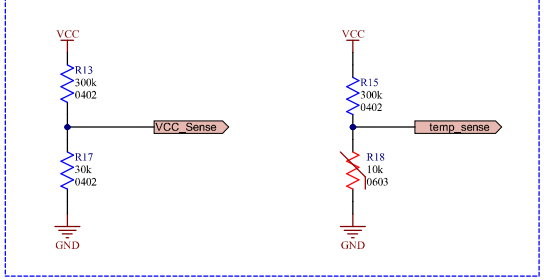
FD-CAN Driver



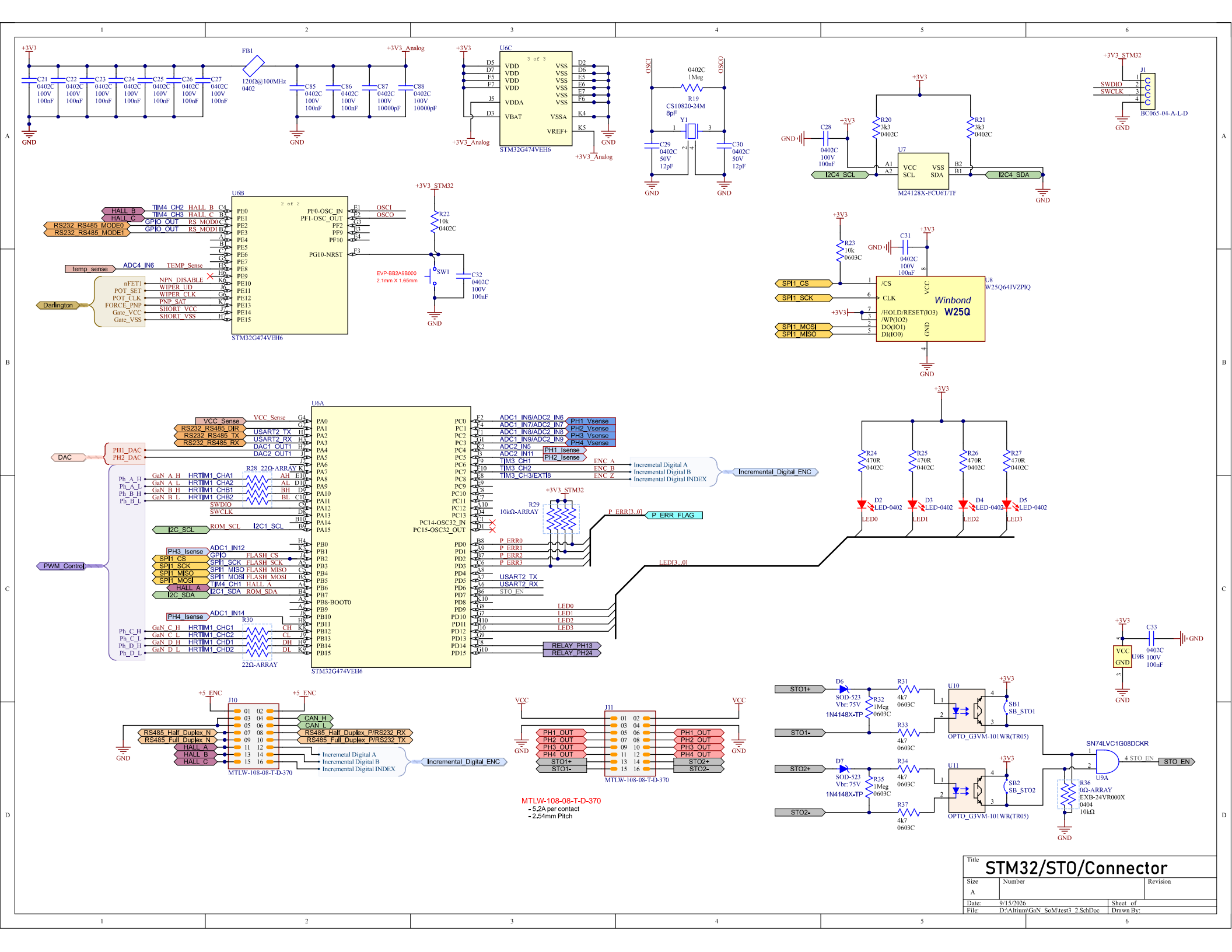
5V Encoder overcurrent Fuse

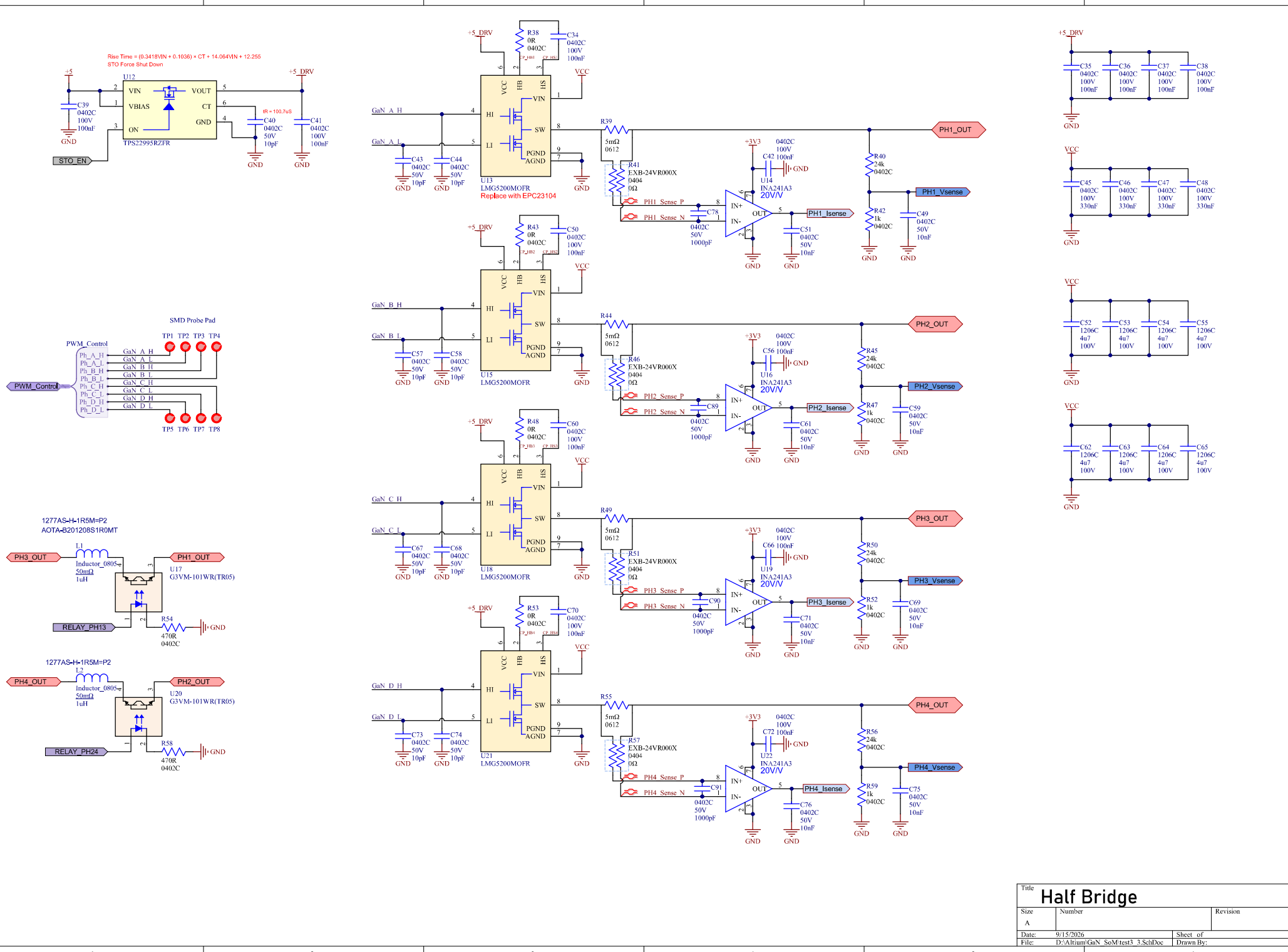


Temperature/Voltage Sense

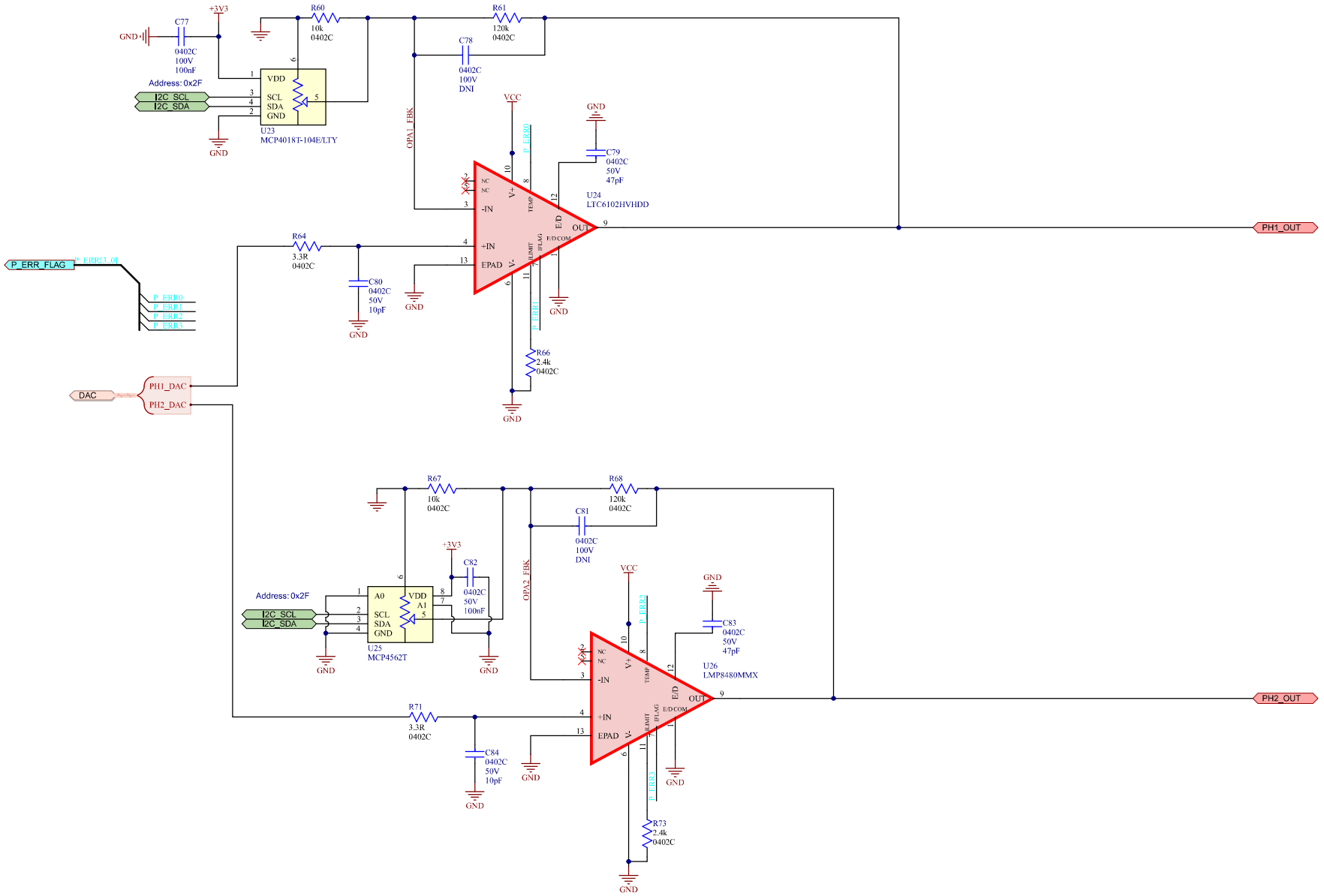


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+3V3
FB1
120Ω@100MHz
0402
+3V3_Analog
C21 0402C 100V 100nF
C22 0402C 100V 100nF
C23 0402C 100V 100nF
C24 0402C 100V 100nF
C25 0402C 100V 100nF
C26 0402C 100V 100nF
C27 0402C 100V 100nF
C85 0402C 100V 100nF
C86 0402C 100V 100nF
C87 0402C 100V 10000pF
C88 0402C 100V 10000pF
GND
U6C
3 of 3
VDD
VSS
VDDA
VBAT
VREF+
STM32G474VEH6
OSCI
OSCO
0402C 1Meg
R19
CS10820-24M 8pF
Y1
C29 0402C 50V 12pF
C30 0402C 50V 12pF
+3V3_STM32
J1
SWDIO
SWCLK
BC065-04-A-L-D
R20 3k3 0402C
R21 3k3 0402C
C28 0402C 100V 100nF
U7
VCC
SCL
SDA
VSS
M24128X-FCU6T/TF
I2C4_SCL
I2C4_SDA
U6B
2 of 2
PF0-OSC_IN
PF1-OSC_OUT
PG10-NRST
R22 10k 0402C
SW1
EVP-BB2A9B000
2.1mm X 1.65mm
C32 0402C 100V 100nF
HALL_B
HALL_C
RS232_RS485_MODE0
RS232_RS485_MODE1
temp_sense
ADC4_IN6
TEMP_Sense
Darlington
nFET1
POT_SET
POT_CLK
FORCE_PNP
Gate_VCC
Gate_VSS
NPN_DISABLE
WIPER_UD
WIPER_CLK
PNP_SAT
SHORT_VCC
SHORT_VSS
R23 10k 0603C
C31 0402C 100V 100nF
U8
W25Q64JVZPIQ
Winbond
W25Q
/CS
CLK
/HOLD/RESET(IO3)
/WP(IO2)
DO(IO1)
DI(IO0)
SPI1_CS
SPI1_SCK
SPI1_MOSI
SPI1_MISO
U6A
VCC_Sense
RS232_RS485_DIR
RS232_RS485_TX
RS232_RS485_RX
USART2_TX
USART2_RX
DAC1_OUT1
DAC2_OUT1
PH1_DAC
PH2_DAC
DAC
R28 22Ω-ARRAY
Ph_A_H
Ph_A_L
Ph_B_H
Ph_B_L
GaN_A_H
GaN_A_L
GaN_B_H
GaN_B_L
HRTIM1_CHA1
HRTIM1_CHA2
HRTIM1_CHB1
HRTIM1_CHB2
SWDIO
SWCLK
I2C_SCL
ROM_SCL
I2C1_SCL
I2C_SDA
ROM_SDA
PWM_Control
PH3_Isense
PH4_Isense
ADC1_IN12
ADC1_IN14
SPI1_CS
SPI1_SCK
SPI1_MISO
SPI1_MOSI
HALL_A
FLASH_CS
FLASH_SCK
FLASH_MISO
FLASH_MOSI
TIM4_CH1
R30
22Ω-ARRAY
Ph_C_H
Ph_C_L
Ph_D_H
Ph_D_L
GaN_C_H
GaN_C_L
GaN_D_H
GaN_D_L
HRTIM1_CHC1
HRTIM1_CHC2
HRTIM1_CHD1
HRTIM1_CHD2
PC14-OSC32_IN
PC15-OSC32_OUT
ADC1_IN6/ADC2_IN6
ADC1_IN7/ADC2_IN7
ADC1_IN8/ADC2_IN8
ADC1_IN9/ADC2_IN9
ADC2_IN5
ADC2_IN11
TIM3_CH1
TIM3_CH2
TIM3_CH3/EXTI8
PH1_Vsense
PH2_Vsense
PH3_Vsense
PH4_Vsense
PH1_Isense
PH2_Isense
ENC_A
ENC_B
ENC_Z
Incremetal Digital A
Incremental Digital B
Incremental Digital INDEX
Incremental_Digital_ENC
R29 10kΩ-ARRAY
P_ERR[3..0]
P_ERR_FLAG
P_ERR0
P_ERR1
P_ERR2
P_ERR3
LED[3..0]
LED0
LED1
LED2
LED3
RELAY_PH13
RELAY_PH24
STO_EN
R24 470R 0402C
R25 470R 0402C
R26 470R 0402C
R27 470R 0402C
D2 LED-0402
D3 LED-0402
D4 LED-0402
D5 LED-0402
C33 0402C 100V 100nF
U9B
VCC
GND
J10
+5_ENC
CAN_H
CAN_L
RS485_Half_Duplex_N
RS485_Full_Duplex_N
RS485_Half_Duplex_P/RS232_RX
RS485_Full_Duplex_P/RS232_TX
MTLW-108-08-T-D-370
J11
PH1_OUT
PH2_OUT
PH3_OUT
PH4_OUT
STO1+
STO1-
STO2+
STO2-
MTLW-108-08-T-D-370
- 5.2A per contact
- 2.54mm Pitch
D6 SOD-523 Vbr: 75V 1N4148X-TP
R31 4k7 0603C
R32 1Meg 0603C
R33 4k7 0603C
U10
OPTO_G3VM-101WR(TR05)
SB1
SB_STO1
D7 SOD-523 Vbr: 75V 1N4148X-TP
R34 4k7 0603C
R35 1Meg 0603C
R37 4k7 0603C
U11
OPTO_G3VM-101WR(TR05)
SB2
SB_STO2
SN74LVC1G08DCKR
U9A
R36 0Ω-ARRAY EXB-24VR000X 0404 10kΩ
Title
STM32/STO/Connector
Size
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Number
Revision
Date: 9/15/2026
File: D:\Altium\GaN_SoM\test3_2.SchDoc
Sheet_of
Drawn By:

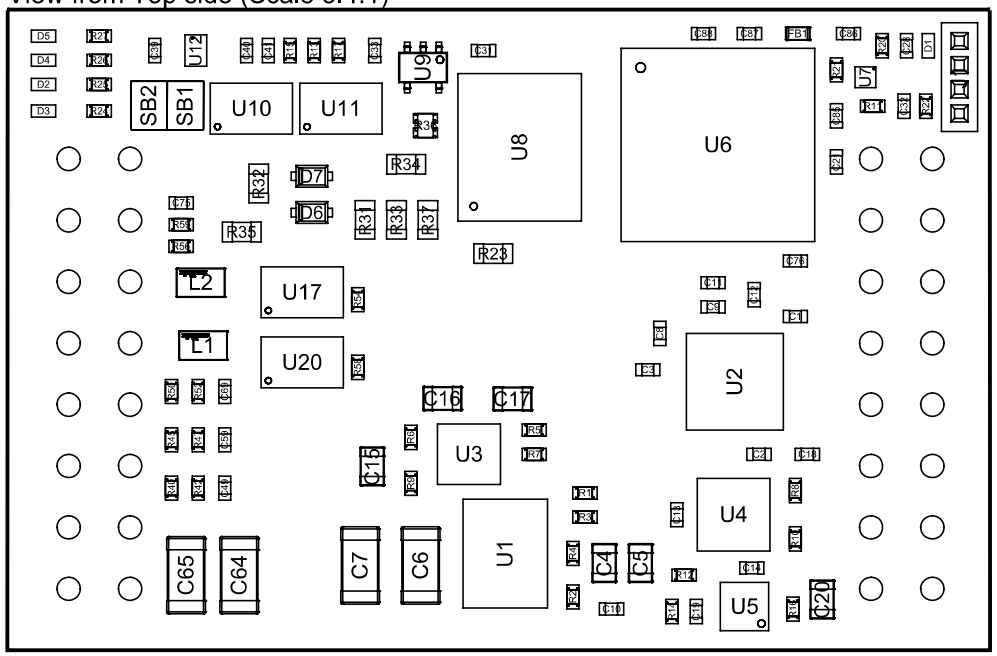


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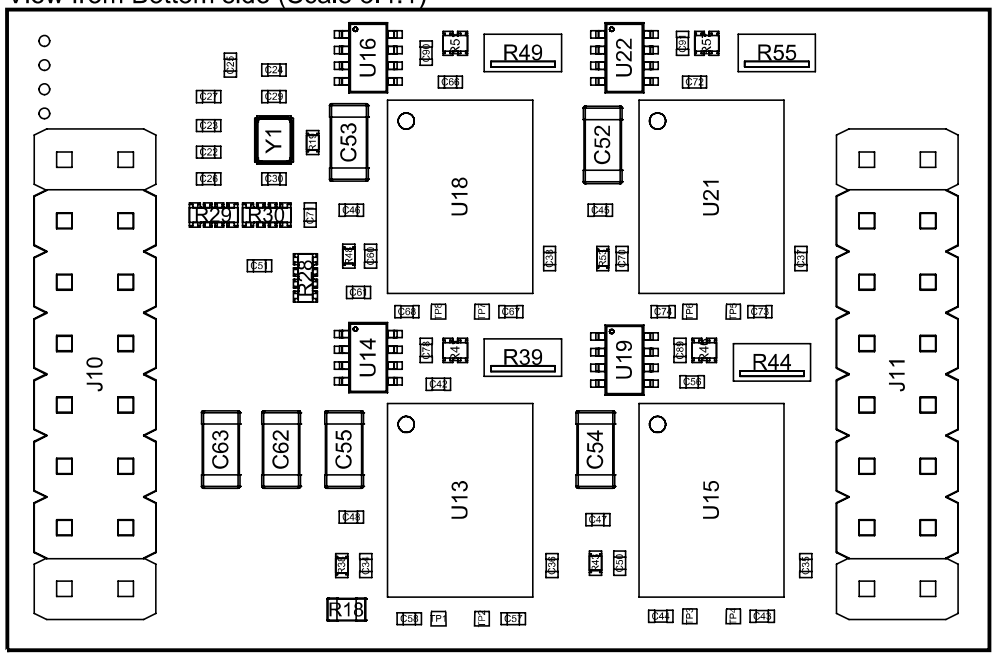


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Date: 9/15/2026		Sheet of
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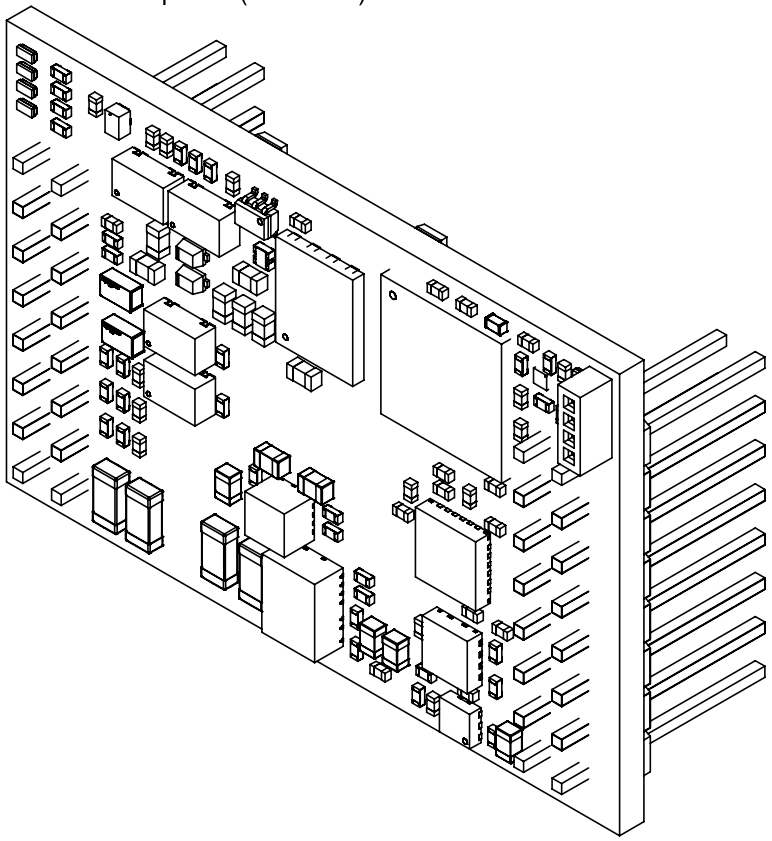
View from Top side (Scale 3.4:1)



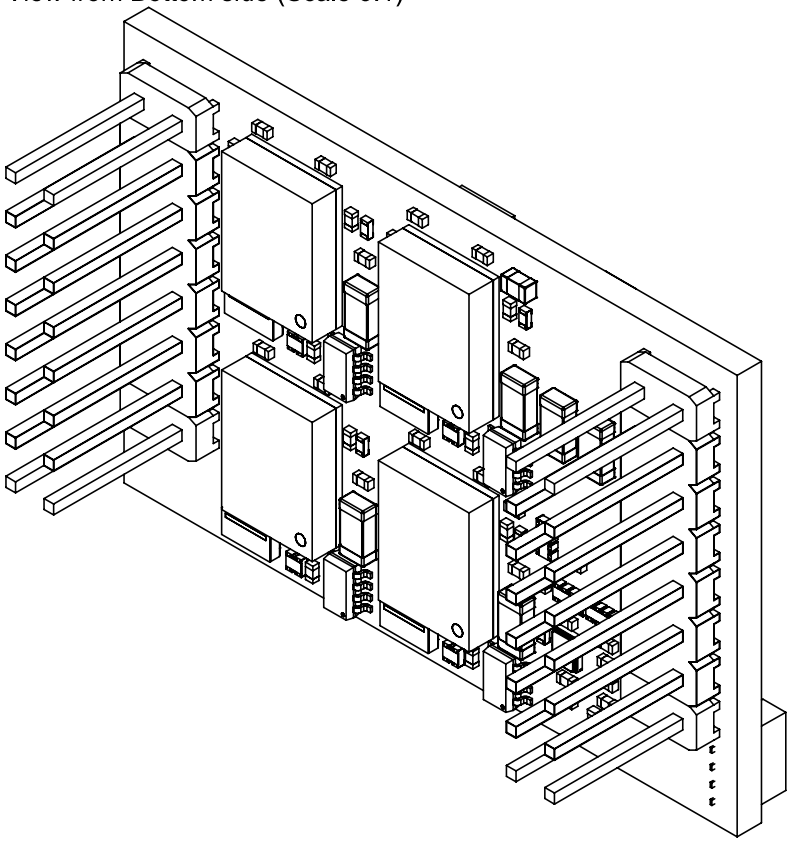
View from Bottom side (Scale 3.4:1)



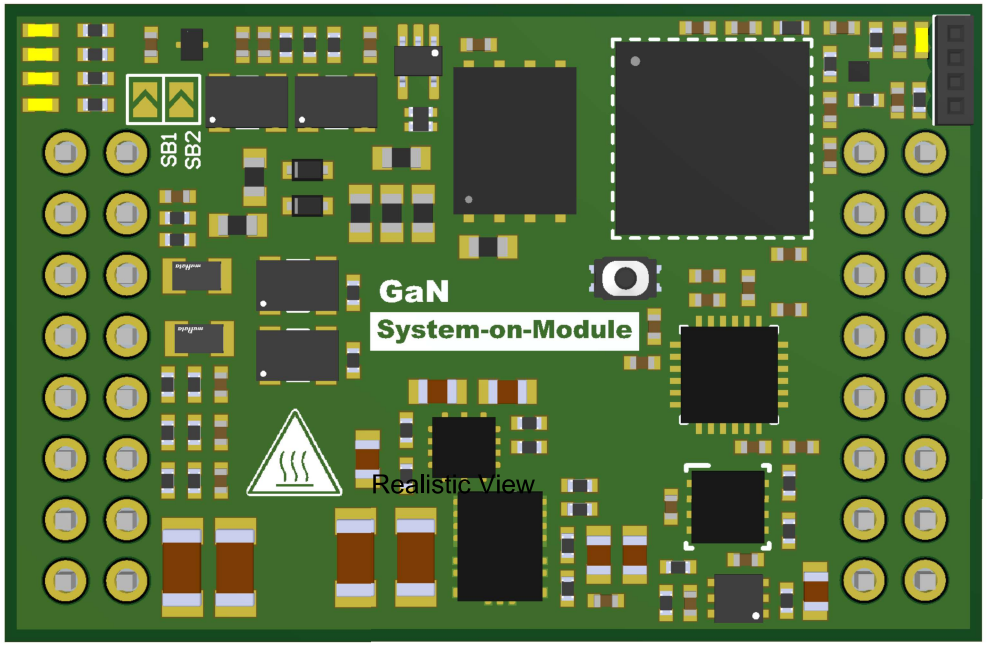
View from Top side (Scale 3:1)



View from Bottom side (Scale 3:1)



Realistic View



Realistic View

